

mounted on the PCBAs, still tested separately due to their nature).

4. Power supplies and battery systems used.

In addition to testing in the manufacturing line, designers have to test the product before the design is moved to the manufacturing line. These tests fall into two main categories:

1. Design verification and validation tests.

2. Compliance/certification tests which are typically one-time and done on a specific number of samples. Special code called field support firmware enables support engineers diagnose the system in the field to isolate the problem unit.

Types of embedded systems testing

When an embedded system is designed, designers need to understand the impact of testing on the product throughout its life. Embedded system testing is split into two major categories.

1. Tests that are done once or twice during the design phase. These tests are controlled by the design team and are mainly firmware-focused. The design phase tests are non-recurring tests used by the designer to validate the system.

2. Tests that are done in the manufacturing line. Every product is subjected to the tests and the number of tests depend on the product complexity and volume.

Non-recurring tests

Non-recurring tests are carried out on the prototypes and pre-release manufactured items to ensure the systems meet all the functionalities. There are two types of tests under this category.

1. Firmware based solution which is typically developed by the designers to initially bring up the newly designed and assembled PCBAs. While this software typically helps to test all the CPU-accessible peripherals only, it is normally used to bring up the board and also validate the parameters of the PCBA.

2. Tests that are conducted (non-recurring) to meet emission, safety, and environmental standards of different countries/zones where the products will be sold. This may involve develop-

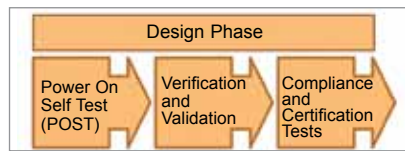


Fig. 2: Design phase

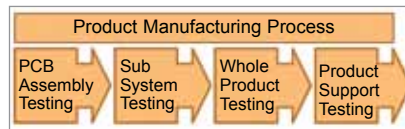


Fig. 3: Different tests that are carried out in the manufacturing line

ing a special firmware to maximise the emissions for compliance testing.

Recurring tests

Recurring tests are typically carried out in the manufacturing line and on every piece of embedded system that is assembled and shipped out. Fig. 3 shows the three different recurring tests at the manufacturing end—PCB assembly testing, sub-system testing, and whole product testing.

The fourth test, product support testing, is typically loaded into the system but will be used only by the field support team to diagnose the problems when the customer reports a malfunction. In the manufacturing line, this test is executed to ensure that the system is able to detect the components properly.

Now that we have seen the types of tests, let us see brief details of the tests.

Understanding the tests

Fig. 4 depicts high-level development flow of embedded systems and the tests that need to be done in each of these phases. The key to understanding the tests depends on the design of the embedded system.

If an embedded system must be tested close to hundred percent, it is vital that testability be designed into the system right from the beginning. Planning to define testing after the design of the system is suicidal. Testability should be a must-have feature or requirement and not an optional one.

Designing a testable embedded system is essential for a successful product. Let us first see the different tests that have to be done before the embedded system is shipped out.

Fig. 4 shows that there are seven types of tests to be done in the entire

product lifecycle. Each of them has a different purpose and has to be done differently.

A classic embedded system has seven tests in its full lifecycle in two phases. The first phase is the design phase where the embedded system takes the final shape from concept to final product. However, at the end of the phase, the product will be ready for transition to manufacturing after design verification and validation.

The tests carried out in the design phase are focused on validating the embedded system for its performance and functionality. The design phase tests are typically one-time. However, the firmware developed for these tests will be used in the product subsequently for testing in manufacturing process and to detect any fault in its working during its life.

There are three types of tests that need to be done, but a well-designed code base can easily be customised to do the three tests that need to be done in the design phase. In the design phase, no special testers are developed, and only dedicated software/firmware are developed as design verification and validation needs sophisticated test equipment to validate the system to ensure the design parameters are validated.

Manufacturing tests address a different type of need. The manufacturing testers' main function is to test every manufactured product completely, in a short time and at a low cost. In addition, the people who operate are technicians and not design engineers. Thus, the tests should be simple and deliver consistent results throughout the life of the product's production.

Design phase tests

As we have seen in Fig. 4, there are three design phase tests that need to be done before the product can be moved to the manufacturing line for large-volume manufacturing. Design-phase tests are more programming intensive. The hardware has to be tested with calibrated test-and-measurement instruments for verifying the functionality and parameters.

The software/firmware that is developed has distinct uses. Let us see one by one. Since most current-day embedded systems are processor- or